Visual Geolocation in Natural Environments Using Modern Vision Models

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1 Public Repository

The project is publicly available on GitHub at the following link:

`https://github.com/DianaDrzikova/Geolocation`

The repository includes all relevant source code and an evaluation notebook.

2 Task Definition

The goal of visual geolocation is to estimate the GPS coordinates of a given photograph purely from its visual content. The special focus is on natural outdoor scenes, for which standard approaches (e.g., those trained primarily on urban data) often struggle due to the lack of distinctive human-made features.

Our objective in this project is to evaluate how well **modern vision models**, such as the **Osv5M model**, can perform geolocation on the **CrossLocate dataset** of natural scenes. We will investigate both regression-based approaches (directly predicting the latitude and longitude of a single photograph of a natural landscape) and classification-based approaches (e.g., clustering into discrete cells). We plan to refine (fine-tune) these models on the CrossLocate training subset to determine whether specialized training in natural scenery improves results.

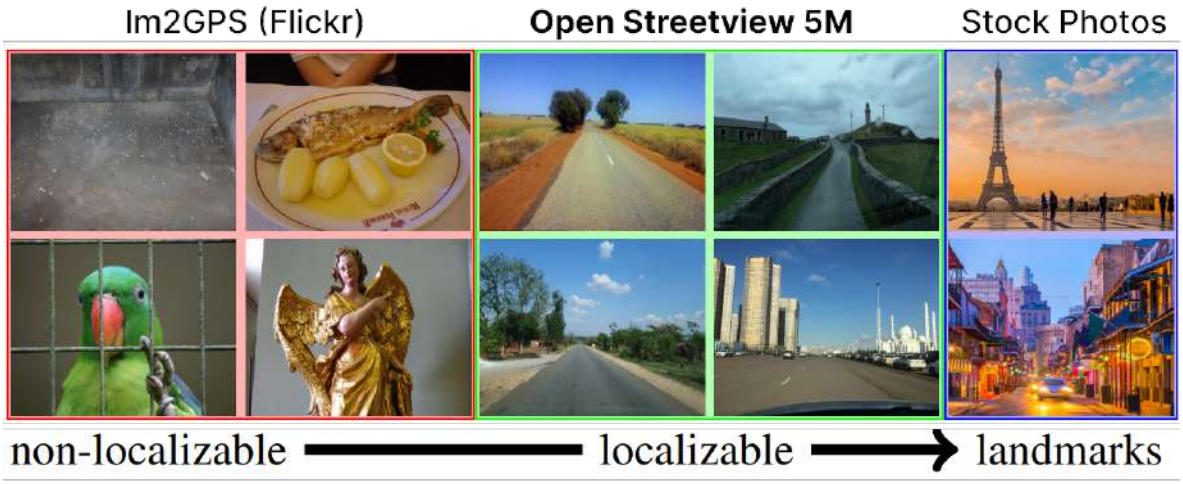


Figure 1: Examples of images with varying localizability. Some images, such as iconic landmarks (right), are easily localizable, while others (left), like food or animals, lack geographic cues. OpenStreetView 5M (center) contains moderately localizable natural scenes. The image was adapted from the Osv5M paper.

3 Overview of Existing Methods

We will work with two key open source projects: **OpenStreetView-5M**[1] and **CrossLocate**[2].

3.1 CrossLocate

CrossLocate [2] is a cross-modal geolocalization approach that matches real photographs to synthetic images (e.g., depth maps, silhouette maps, segmentation maps) generated from 3D terrain models. By rendering multiple synthetic views of the same geotagged location, CrossLocate effectively learns to match key terrain-based features in a query photograph to features in its synthetic counterpart. Therefore, if an actual real photograph resembles the depth or silhouette of a rendered viewpoint, it is likely that their corresponding coordinates are close. This cross-modal learning allows for accurate localization in natural environments, where purely urban cues like street signs or buildings are minimal or absent.

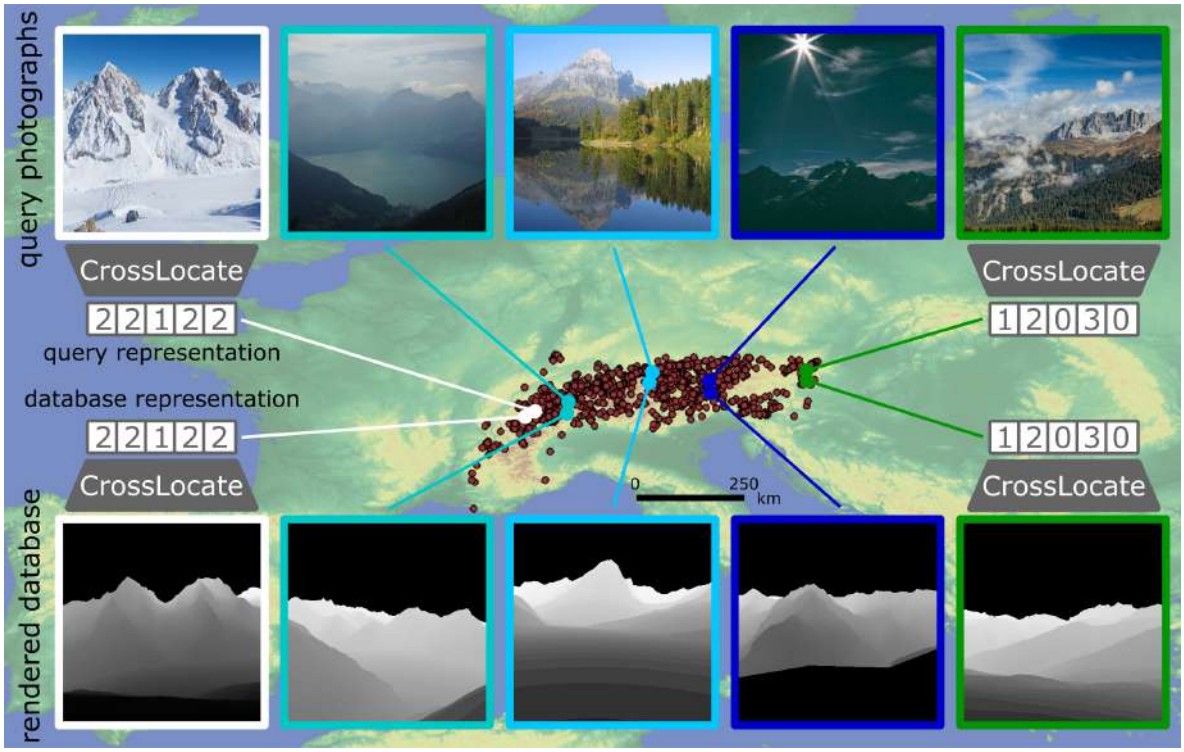


Figure 2: The image demonstrates a basic overview of how the CrossLocate approach works. The descriptor of a query image (nature image) is matched with the descriptor of the modality images (in this case, depth map). The image is adapted from the CrossLocate paper.

[1] https://github.com/gastruc/osv5m
[2] https://github.com/JanTomesek/CrossLocate

3.2 OpenStreetView-5M

Osv5M [1] introduces a massive dataset of about 5.1 million geotagged street view images collected from OpenStreetMap. The dataset is globally distributed, ensuring broad geographic coverage. It was originally designed for large-scale geolocalization tasks, with careful train/test separation to prevent overfitting to specific regions. Models trained on Osv5M learn discriminative visual features for wide-scale geolocalization, but may lack fine-grained terrain-based cues, since Osv5M is often dominated by urban or street-centric images. We are interested in examining how these city-trained or street-trained Osv5M models transfer to natural scenes in mountainous or wilderness areas, as found in CrossLocate.

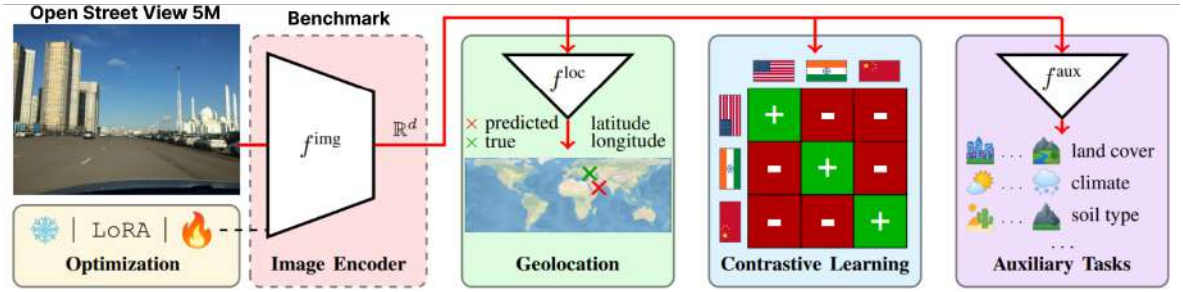


Figure 3: The figure shows the Osv5M model architecture. An input image is encoded by f_{img} and passed to f_{loc} to predict latitude and longitude. Contrastive learning (cyan) and auxiliary tasks (purple) like land cover or climate may be added. The image is adapted from the Osv5M paper.

4 Dataset

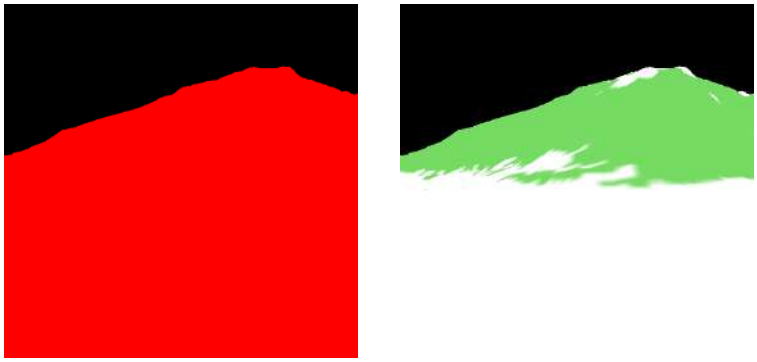
We will be working with the **CrossLocate**[3] dataset, which consists of:

- **Sparse dataset**: Contains $\sim 3,000$ unique real places (photos) and $37,000$ synthetic renderings of different modalities (e.g., depth, silhouettes).
- **Uniform dataset**: Contains $\sim 12,000$ unique real places (photos) and 11M synthetic renderings. This larger set is especially suitable for large-scale experiments.

We focus on the **Uniform dataset**, since the CrossLocate paper reported its final evaluations on that variant. The dataset includes:

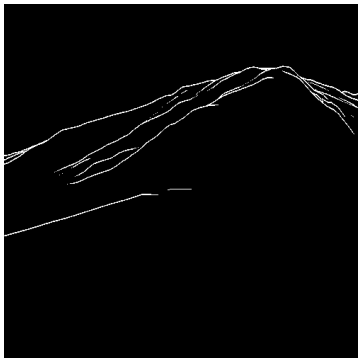
- Real photography of various landscapes (mountains, valleys, etc.).
- Synthetic images (depth, segments, and silhouette) rendered from the same vantage points.
- Complementary CSV and `npy` files with relevant numeric data (latitude, longitude, altitude, etc.).

Figure 4 shows examples of some synthetic modalities (depth, segmentation, silhouette) and the corresponding real photograph from the Sparse subset. The same modalities (only on a much larger scale) exist for the Uniform subset.



(a) Depth map

(b) Segmentation



(c) Silhouette map



(d) Query image

Figure 4: Examples of synthetic modalities and the query photograph from the Sparse CrossLocate dataset.

[3] https://cphoto.fit.vutbr.cz/crosslocate/dataset/datasets/

Table 1 displays an example row from the CSV metadata, illustrating how the dataset encodes key parameters such as position (latitude, longitude), orientation (yaw / pitch / roll) and field of view.

place number	0000001
place name	28488116812_f5a57ca0f6_k
latitude	46.2173
longitude	10.1663
elevation	439.500000
yaw [rad]	0.272811
pitch [rad]	0.243122
roll [rad]	0.0148168
horizontal FOV [rad]	0.549165

Table 1: Example of a single row in the CrossLocate CSV metadata.

We comply with the provided train/validation/test splits of the Uniform dataset to maintain consistency with prior CrossLocate experiments.

5 Evaluation Method

We will evaluate geolocation accuracy using a **distance-based recall** metric. Specifically, for each query image, we obtain the predicted GPS coordinates and compute the geodesic distance to the ground-truth coordinates. We then plot a recall curve that shows, for a range of distance thresholds (e.g., 0 m to 10,000 m), what fraction (recall) of queries falls within each threshold.

5.1 Recall Curve Explanation

A recall value of 10% at a 1 km threshold implies that the model correctly locates 10% of queries (within 1 km of their ground truth). By sweeping from small to large distance thresholds, we get a curve showing how quickly the recall saturates. A steeper curve with higher recall at lower distances indicates a more accurate geo-localization model.

Figure 5 (adapted from the CrossLocate paper) is an example of such a curve. We will replicate and compare similar plots for our chosen models.

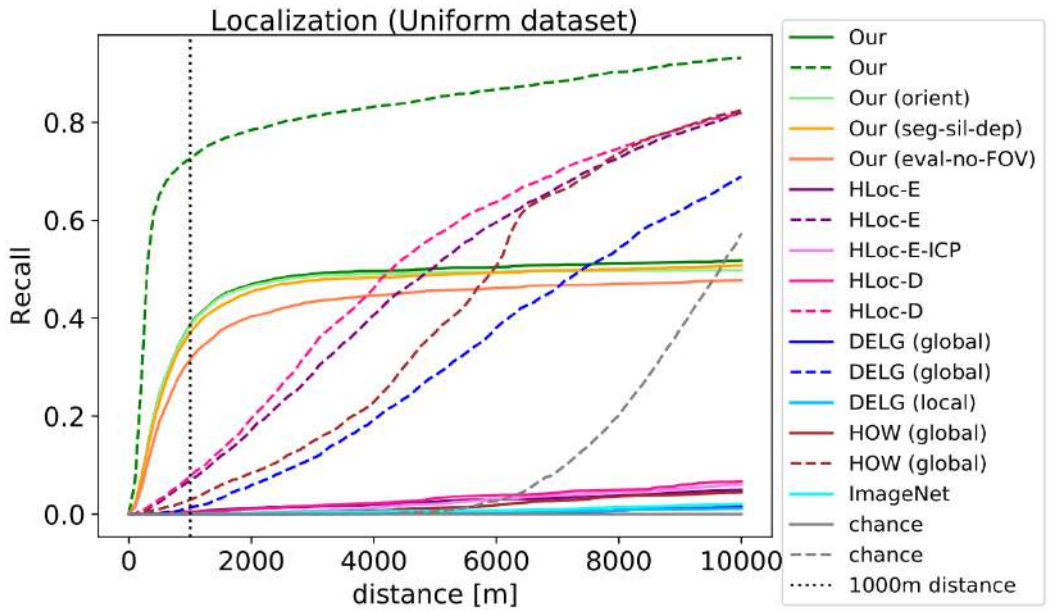


Figure 5: Recall-distance curve from the CrossLocate paper. Higher lines indicate better localization performance at stricter thresholds. The "Our" curves are the results of the authors of the CrossLocate approach.

Additionally, for our planned classification-based approach, we aim to cluster the CrossLocate Uniform dataset into classes (Alps region). We will then measure the accuracy, precision, recall and F1 score of classification against ground-truth classes to assess performance.

6 Baseline Solution and Evaluation

We took the `osv5m/baseline` HuggingFace model–trained originally on the Osv5M dataset and applied it directly to 12,353 query images from the CrossLocate Uniform dataset.

The model outputs the latitude and longitude in radians. Key findings:

- **Processing time:** About **38 minutes** on a NVIDIA GTX 1650 Ti to evaluate all 12,353 queries.
- **Localization error:** Only **3.1%** of queries were localized within 10 km. The geodesic distances showed:
  - **Min distance:** 1,938.16 m
  - **Max distance:** 12,349,954.30 m
  - **Mean distance:** 421,827.08 m
  - **Median distance:** 98,329.44 m

Figure 6 shows two example query images: the one with the best guess (402 m) and the one with the worst guess (over 18,000 km).



(a) Best guess (1,938 m off).



(b) Worst guess (12,349,954 m off).

Figure 6: Examples from the CrossLocate Uniform dataset with large variation in localization error.

As mentioned, the predicted longitude and latitude were in radians, so we converted them to match the ground-truth values in degrees. We computed the geodesic distance (in meters) between the predicted and the ground truth using the `geopy` library.

We then plotted the recall-distance curve (Figure 7) by counting the fraction of queries within each distance threshold. As shown, the recall saturates only at around 3%, indicating that the Osv5M model does not transfer well to these natural mountain images.

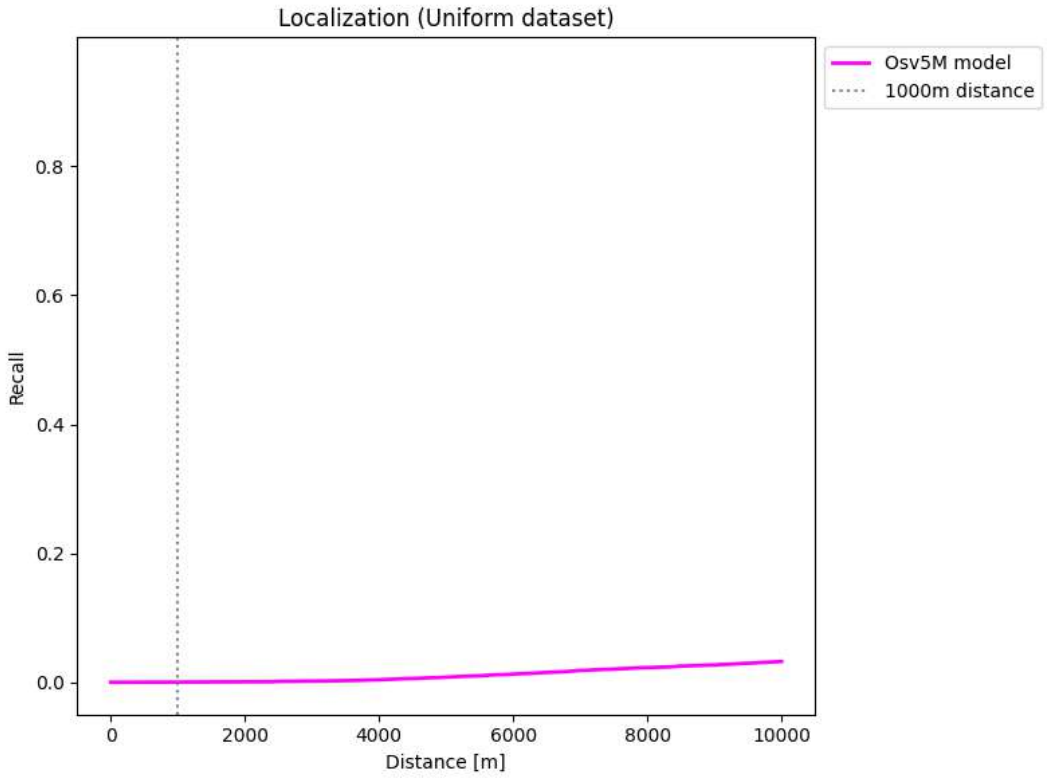


Figure 7: Recall vs. distance for the Osv5M baseline on the CrossLocate Uniform dataset. Maximum recall $\approx 3\%$.

This baseline illustrates the difficulty of localizing purely natural landscapes using an urban-biased model. We propose to fine-tune the same model on CrossLocate data to improve these results.

7 Project Plan

The following list shows an overview of our work. Details will be explained in the next section.

- **Baseline evaluation** We have tested an existing baseline Osv5M model on the Cross-Locate Uniform dataset and computed recall-distance curves.
- **Annotations** We have implemented various annotation strategies in the dataset.
  - K-means clustering
  - Clustering based on OpenStreetMap (naive and balanced)
  - Optics clustering (naive and balanced)
- **Regression training** We have tried various fine-tuning strategies with the Osv5M model:
  - New pure regression head with Osv5M backbone (full, freeze)
  - Original hybrid head of Osv5M (full, freeze)
  - Supervised regression of Osv5M model for admin2 region and country, admin1, admin2 regions (full)
  - Fine-tuning regression on the simple CLIP model and the GeoClip model
- **Classification/Hybrid** We have trained Osv5M model for classification (country, admin1 region)
- **Evaluation** We have extensively evaluated the performance of all models with suitable evaluation metrics.

For both classification/hybrid and regression models, we selected the model that performed the best during the training for a chosen number of epochs. We also often use the term **Before** for the **Baseline** Osv5M model.

8 Implementation details

8.1 Annotations

The original CrossLocate dataset was uniformly sampled but lacked any class annotations, which posed a challenge for training classification-based models. To address this, we implemented several annotation strategies to assign class labels based on geospatial information.

8.1.1 Clustering methods

We experimented with three main clustering approaches to derive class labels from GPS coordinates. For more outputs, see `eval_classes.ipynb`.

K-Means Clustering:
We applied standard K-Means with 50 clusters. However, this approach did not provide a balanced distribution of samples across clusters, as visualized in the Figure 8. The centroids had good spatial distribution over the Alps but the distribution between classes was poor. Many clusters had too few or too many samples, making them unsuitable for classification (Figure 9).

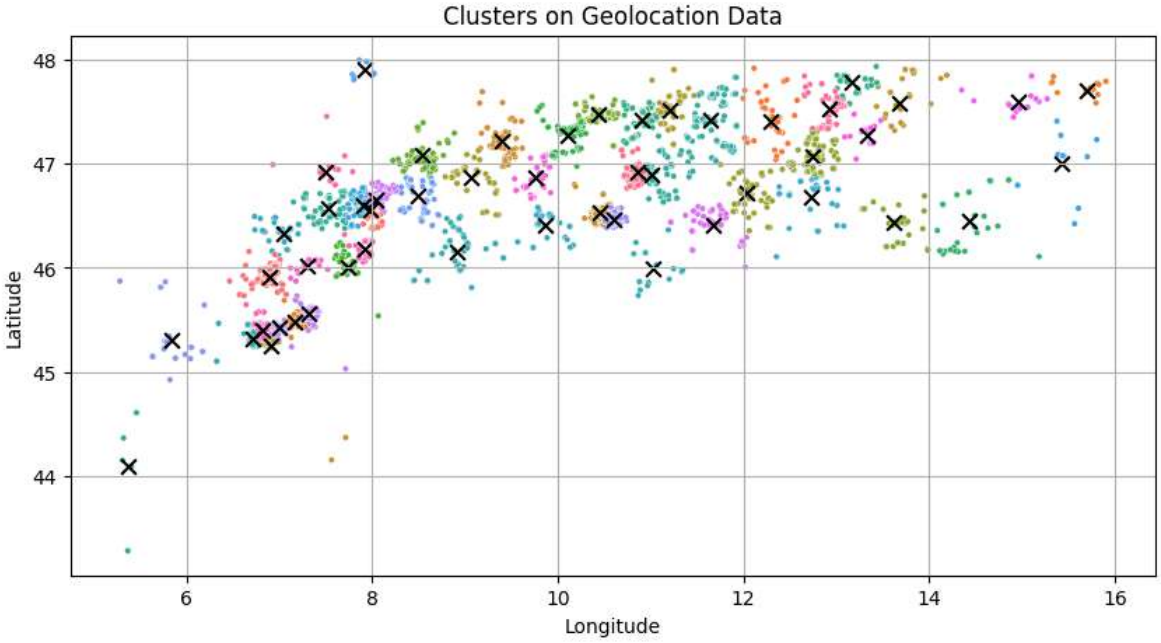


Figure 8: K-means centroids

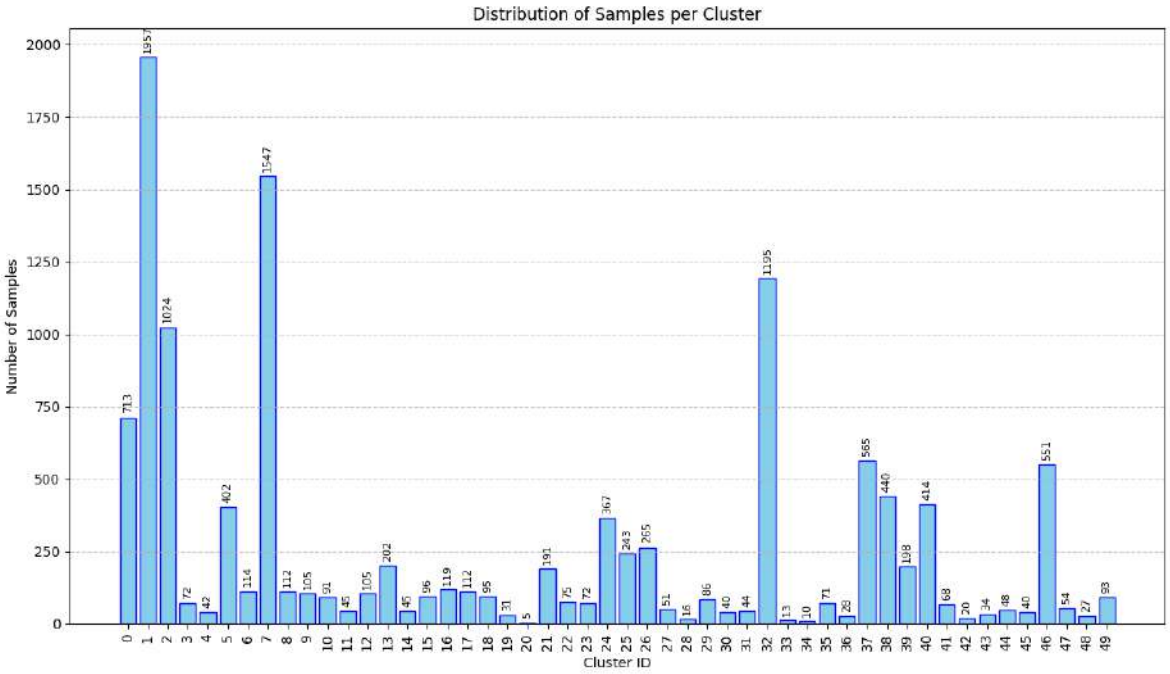


Figure 9: K-means distribution

OpenStreetMap-based Annotation:

Using reverse geocoding via OpenStreetMap, we assigned each image to hierarchical administrative regions:

- Country level: 7 classes
- Admin1 (e.g., states/provinces): 58 classes
- Admin2 (e.g., counties/districts): 127 classes

Figures 12 and 11 show the same distribution of classes, the same way as K-means.

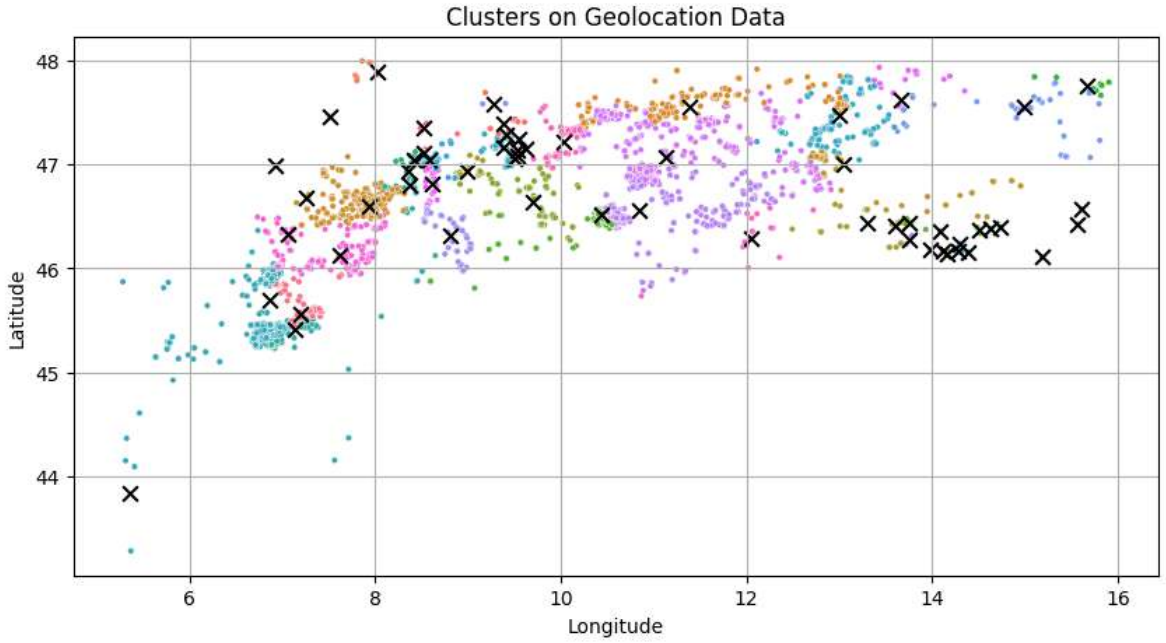


Figure 10: Admin1 centroids

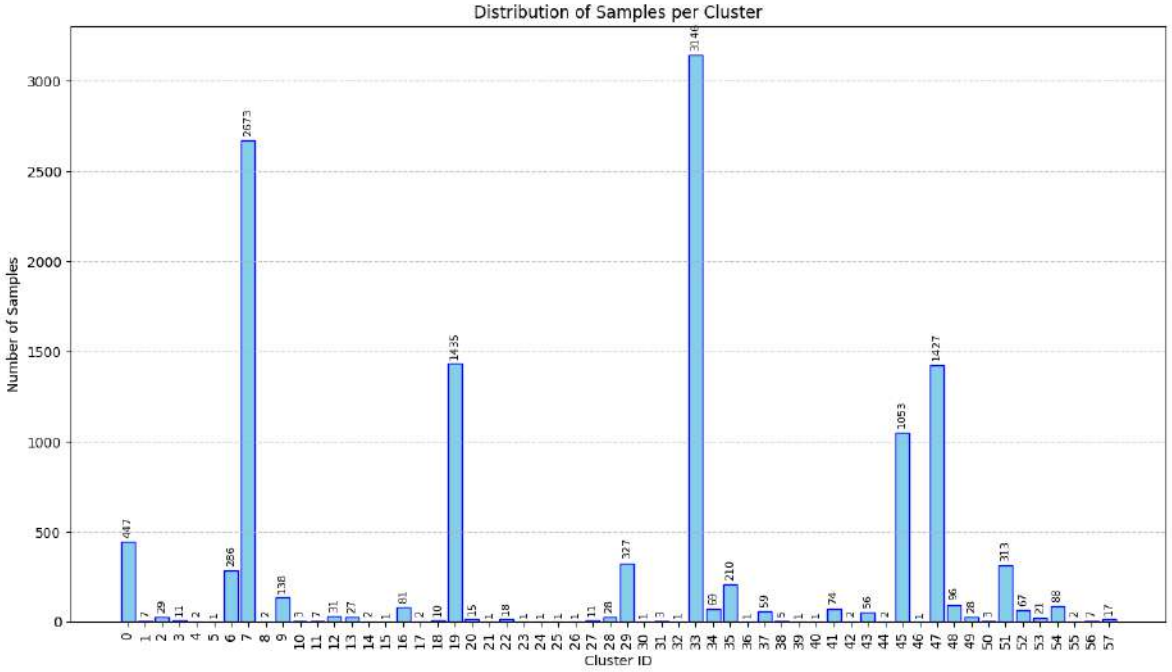


Figure 11: Admin1 distribution

We also considered using cities as classes, but the number of unique city labels was prohibitively high, resulting in an overly fragmented class distribution. City-level classification was discarded.

OPTICS Clustering (PIGEON-inspired):
Inspired by the PIGEON approach[4], we experimented with OPTICS clustering to define spatial classes. This method was designed to mimic administrative boundaries while keeping cluster sizes manageable. In this approach:

- Class labels are derived from spatial density-based clusters.
- Noise points are reassigned to the nearest cluster using a KD-tree.
- A maximum cluster size constraint of 300 samples was enforced.

8.1.2 Balancing OpenStreetMap Clusters

Administrative divisions from OSM (especially admin1 and admin2) exhibited a highly skewed distribution, with some regions having hundreds or thousands of samples and others having very few.

To mitigate this, we implemented a balancing strategy:

- **Splitting large regions:** Regions with more than 300 samples were split into smaller subclusters using recursive K-Means.
- **Merging small regions:** Regions with fewer than 100 samples were merged with nearby larger regions based on haversine distance.
- These transformations ensured that each class contained between 100 and 300 samples, optimizing the dataset for classification training and evaluation.

We provide all annotations in the submitted solution. Because of the specified train/val/test distribution, we worked with pure OpenStreetMap annotations.

[4]https://lukashaas.github.io/PIGEON-CVPR24/

8.2 Regression training

To explore different approaches to geolocation prediction, we trained multiple variants of the Osv5M model and additional CLIP-based models. The following section outlines the training strategies we adopted for both regression tasks. There were only little changes in training parameters as we aimed to get relevant results.

We trained all Osv5M models with a batch size of 16 and a learning rate of $1e-5$. The new head regression was the only model trained with haversine loss, otherwise, we used L1 loss.

We trained the CLIP models with a batch size of 32 and a learning rate of $1e-3$ by which we retrained the Osv5M with these parameters and haversine loss.

8.2.1 Pure Regression with Osv5M

We replaced the original hybrid head of the Osv5M model with a new pure regression head to directly predict GPS coordinates (latitude, longitude). This was motivated by the desire to simplify the architecture and focus solely on regression performance. The training was performed under two modes:

- **Frozen backbone:** Only the regression head was updated.
- **Full training:** The entire model, including the backbone, was fine-tuned.

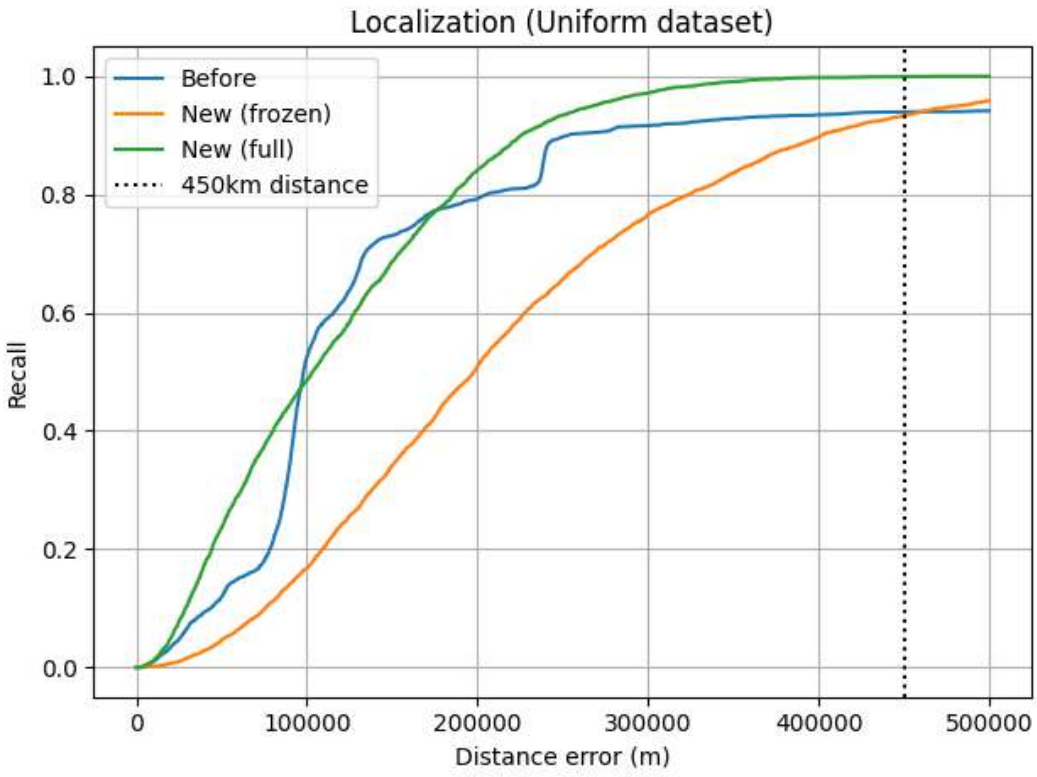


Figure 12: Performance of new pure regression head of Osv5M model.

The fully trained model with L1 provides the most smooth and balanced results outperforming the baseline model. It is able to predict coordinates of images within 450km with 100% success.

Distance Threshold	New (frozen)	New (full)	Baseline
Recall @ 1 km	0.000	0.000	0.000
Recall @ 25 km	0.009	0.078	0.047
Recall @ 200 km	0.509	0.841	0.793
Recall @ 450 km	0.933	0.999	0.940
Recall @ 750 km	0.993	1.000	0.944

Table 2: Recall@Distance comparison for different training modes of the Osv5M model with new regression head.

8.2.2 Hybrid Regression

After successfully extracting the geocells defined by the Osv5M authors, we trained the original hybrid head architecture. We used extracted geocells as input to the model to retrieve GPS coordinates to continue training for regression. This dual-headed approach aligns with the original Osv5M design and can serve for closer comparison for baseline and trained models. We can come to the conclusion that this method of training with full mode degraded the model. On the other hand, the fine-tune mode for the regression head did not change the performance of the baseline model.

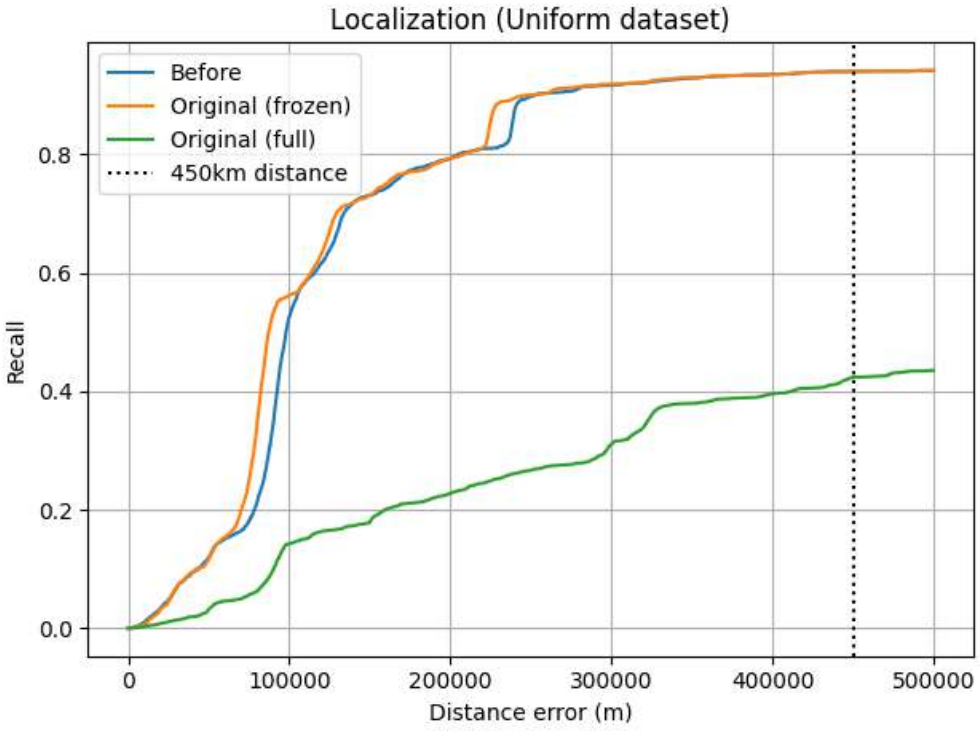


Figure 13: Recall@Distance Performance of finetuned Osv5M model with original hybrid head.

Distance Threshold	Original (frozen)	Original (full)	Baseline
Recall @ 1 km	0.000	0.000	0.000
Recall @ 25 km	0.045	0.011	0.047
Recall @ 200 km	0.793	0.227	0.793
Recall @ 450 km	0.940	0.424	0.940
Recall @ 750 km	0.994	0.445	0.944

Table 3: Recall@Distance comparison for different training modes of the OSV5M model with hybrid head.

8.2.3 Custom Hybrid Regression Using OpenStreetMap Labels

We experimented with an alternative to the OSV5M hybrid head by implementing a custom regression setup where:

- The model outputs a fixed number of coordinates (one per class).
- The correct output is selected using the ground truth class index (e.g., admin2_id or country-admin1-admin2 combination).

This approach essentially performs classification-guided regression. However, results showed performance degradation.

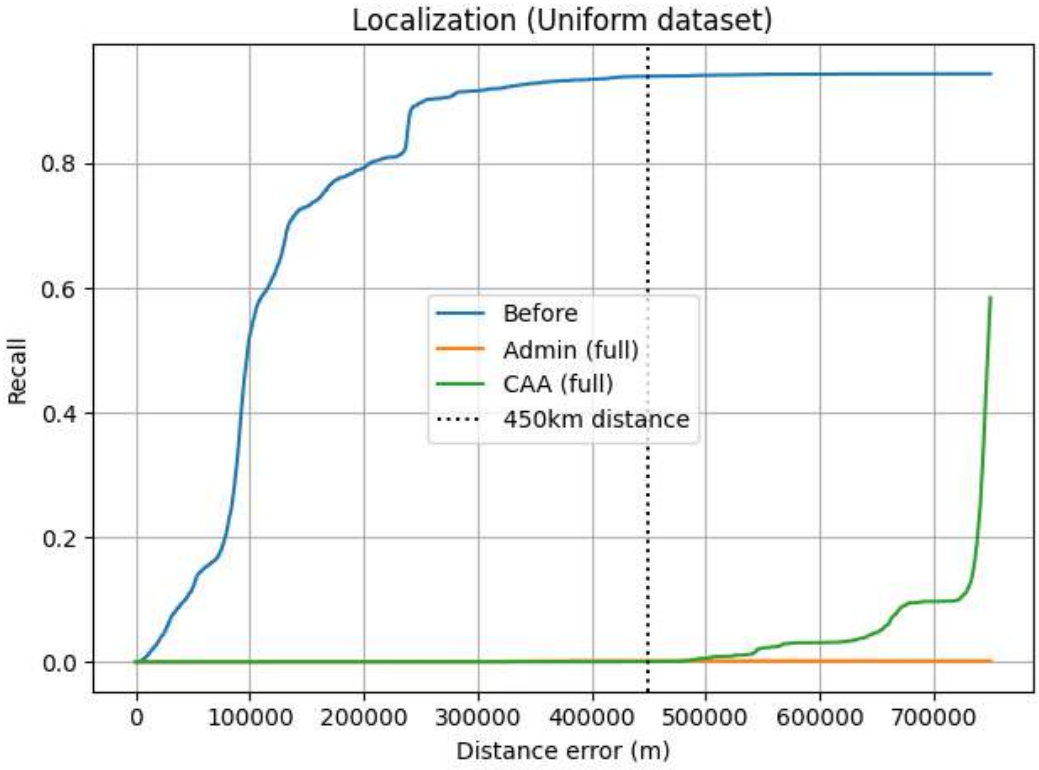


Figure 14: Recall@Distance Supervised regression on Osv5M model.

8.2.4 Clip Models

In addition to Osv5M-based models, we trained a simple CLIP-based model with the architecture of `ViT-L-14` that was pretrained on `laion2b_s32b_b82k` dataset. We trained the GeoCLIP model as well.

- **BasicClip**[5]**:** model developed by **OpenAI** uses the ViT-L/14 backbone (Vision Transformer Large with 14×14 patch size), pretrained on the LAION-2B dataset—a large-scale collection of 2 billion image-text pairs. The specific checkpoint laion2b_s32b_b82k refers to a training run with 32 billion training steps and 82,000 batch size, optimized for broad visual-text alignment. It produces 512-dimensional embeddings suitable for downstream tasks like image retrieval, captioning, and geolocation via transfer learning.
- **GeoClip**[6]**:** model proposed by **Vicente Vivan et al.**, is a contrastive model inspired by CLIP, trained to learn a shared embedding space between images and geographic coordinates. It was trained on 4.7 million geotagged photos (MP-16 dataset), using a ViT-L/14 backbone to generate 512-dimensional image embeddings that can be directly compared to 512-dimensional GPS coordinate embeddings encoded via random Fourier features. This enables zero-shot geolocation by matching visual features to geographic locations.

Each model was trained with the freeze mode, where we trained only a new regression head. We evaluated the performance with our new Haversine-trained head regression model (in the table 4 it will be called Osv5M). We can observe that BasicClip managed to get almost 100% predictions within $350km$, which is a $100km$ decrease of our so far best model (fully trained new regression head Osv5M).

[5]https://github.com/OpenAI/CLIP
[6]https://github.com/VicenteVivan/geo-clip

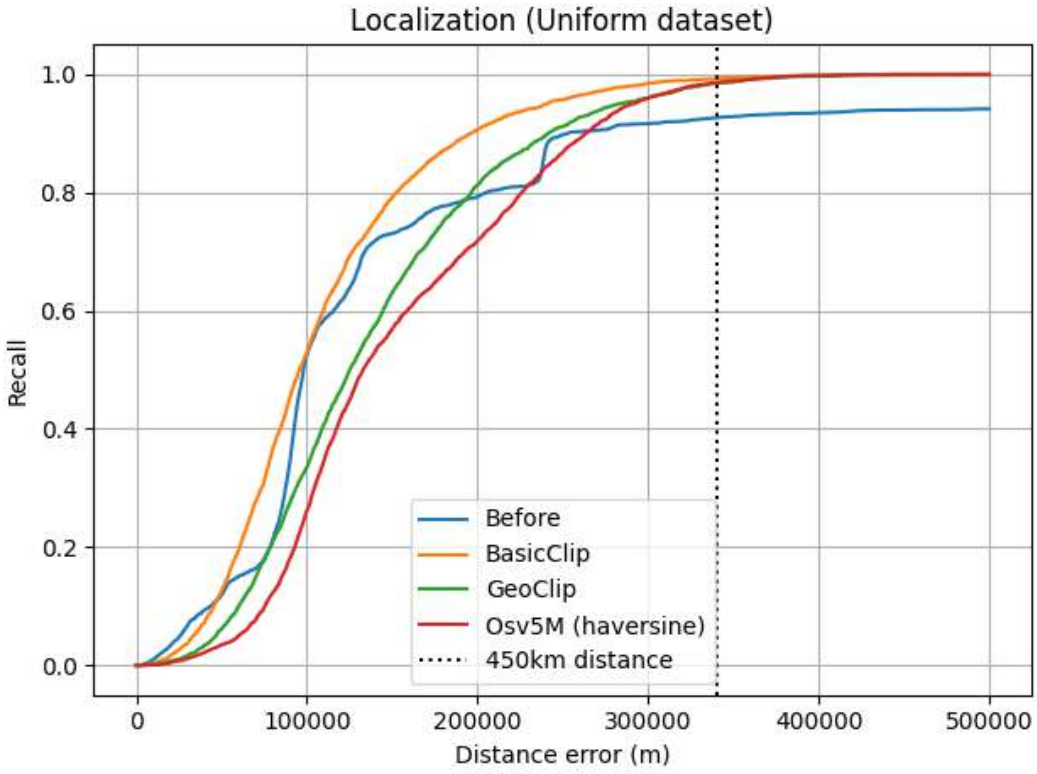


Figure 15: Backbone model comparison.

Distance Threshold	**Basic CLIP**	**GeoCLIP**	**Osv5M**	**Baseline**
Recall @ 1 km	0.000	0.000	0.000	0.000
Recall @ 25 km	0.026	0.007	0.008	0.047
Recall @ 200 km	0.906	0.676	0.718	0.793
Recall @ 350 km	0.994	0.979	0.989	0.940
Recall @ 450 km	0.999	0.998	1.000	0.944

Table 4: Recall@Distance for Basic CLIP, GeoClip, and Osv5M (new regression head using haversine loss) models.

Following table demonstrates evaluation metrics and some other parameters about the models:

Model	**Architecture**	**GeoScore**	**Mean Distance**
BasicCLIP	ViT-L/14 + reg.head	4647.06	141km
GeoCLIP	ViT-L/14 + reg.head	4556.94	155km
OSV5M Haversine	ViT-B-32 + reg.head	4512.14	110km
Baseline	ViT-B-32	4392.87	422 km

Table 5: Clip models, Osv5M Haversine model and Baseline model comparison for Geoscore and mean distance.

Geoscore was computed as in the Osv5M paper: $geoscore = 5000 \exp(-\delta/1492.7)$.

8.2.5 All models for regression

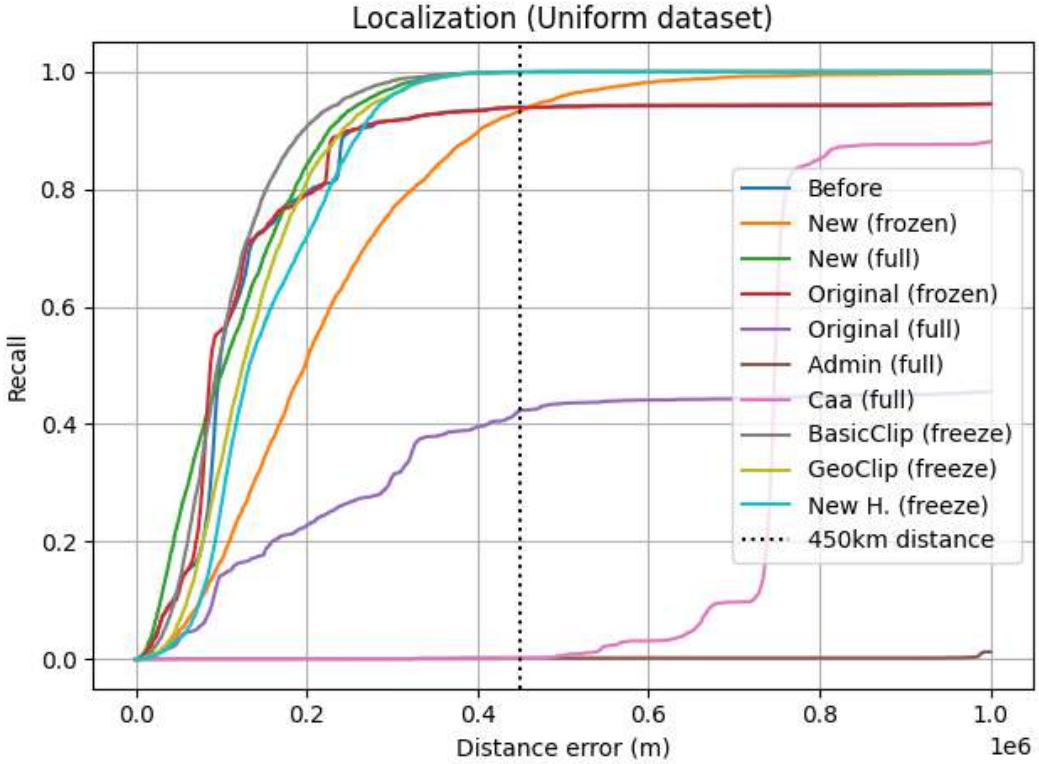


Figure 16: Recall@Distance for all models.

8.3 Classification training

We originally planned to train simultaneously on both regression and classification, but the results were extremely poor, so we decided to reuse the fine-tuned pure regression Osv5M model and trained classification heads for **admin1** and **country**.

Admin1 classification was chosen over admin2 due to a more balanced distribution across train/val/test splits. However, performance was limited:

- The admin1 classifier struggled due to significant class imbalance.
- The country classifier faced an additional issue: the test set consisted predominantly of samples from a country not present in the training set, leading to poor generalization.

We used weighted cross entropy to balance the uneven distribution. We firstly trained without the weights of classes in the dataset, but the results were extremely poor.

8.3.1 Country classification

The main issue with country classification training was that the original train/test/val distribution was selected so the GPS coordinates were not to close to each other, and one country was completely discarded from the training dataset and was present only in the test dataset. We decided to evaluate the performance on a more balanced set of images that is a combination of validation images and test images, where we evaluated 138 images with the class distribution:

Country ID	**Count (GT)**	**Country ID**	**Count (Pred)**
2	44	0	76
0	35	4	40
1	35	3	22
4	24	–	–

Table 6: Comparison of ground truth country distribution and predicted class distribution.

It is not the ideal situation, but otherwise we would get 0% accuracy so we optimized the split in order to show that the model can classify between classes that it was trained on. We got 20.29% accuracy with this distribution.

8.3.2 Admin1 classification

Training classifier on admin1 was even more of an obstacle, as there were 58 different classes with various distributions. We did not balance the evaluation dataset, as the admin1 train/test/val split was more reasonable for the country classes split.

In conclustion to classification training, this was a big obstacle as the original train/test/val split was aimed to train regression for GPS coordinates. Our main condition for training was that we wanted the best geological representation, but in the future we propose to train on balanced datasets even though the geological distribution is skewed.

References

[1] Astruc, G., Dufour, N., Siglidis, I., Aronssohn, C., Bouia, N., Fu, S., Loiseau, R., Nguyen, V. N., Raude, C., Vincent, E., XU, L., Zhou, H., and Landrieu, L. Openstreetview-5m: The many roads to global visual geolocation, 2024.

[2] Tomešek, J., Čadík, M., and Brejcha, J. Crosslocate: Cross-modal large-scale visual geo-localization in natural environments using rendered modalities. In *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (2022), pp. 2193–2202.